

Zofia Marciniak

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ABOUT

Human-Computer Interaction researcher with interest in computational craft, embodied interaction, and creative technology tools. Experienced in sensing systems, fabrication workflows, and user-centered research methods.

SKILLS

Programming	Python, C++, JavaScript, C# with Rhino/Grasshopper
Interactive Systems	Arduino, Raspberry Pi, IMU sensors, motors, custom sensing systems
Design Research	User studies, workshops, thematic coding and analysis
Fabrication	3D printing, laser cutting, computational making, 3D modeling

WORK EXPERIENCE

Engineer, Samsung Electronics September 2025 – July 2026

Developed software for sensing technologies under the Visual Display division.
Part of the HCI Solutions team for hardware platform.

Software Engineering Intern, NoMagic.AI June 2022 – September 2022

Designed a restructure of the robotic arm main thread.
Implemented new Pick-and-Place features and supervised deployment.
Technologies: Python, Robotics.

Software Engineering Intern, Google July 2021 – September 2021

Developed a language parser for internal developer tools in Go.

EDUCATION

August 2026 – present	PhD in Computer Science, ETH Zürich , Switzerland Supervisor: Prof. April Yi Wang Researching Embodied Interactions solutions for AI literacy education
August 2023 – August 2025	M.Sc. Industrial Design, KAIST , Daejeon Thesis: <i>Facilitating In-Situ Creative Exploration and Pattern Sharing in Crochet With Real-Time Stitch Detection and Automatic Pattern Formatting</i> Supervisor: Prof. Andrea Bianchi Secretary, KAIST International Student Association; Graduate Admissions Student Ambassador
September 2019 – August 2023	B.Sc. Double Major in Industrial Design and Computer Science, KAIST , Daejeon Excellence in Leadership Award

RESEARCH EXPERIENCE

Graduate Research Assistant, PEACH LAB, ETH Zürich	Aug 2026 – present
Graduate Research Assistant, Make Lab, KAIST	Aug 2023 – Aug 2025
Led master's thesis research on computational craft support. Developed real-time stitch detection systems for crochet pattern sharing and creative exploration.	
Research Intern, Make Lab, KAIST	Sep 2022 – Aug 2023
Research Intern, HCI Tech Lab, KAIST	Jan 2022 – Jun 2022
Research Intern, CDSN Lab, KAIST	Jul 2020 – Dec 2020

PUBLICATIONS

- Marciniak, Zofia**, Punn Lertjaturaphat, and Andrea Bianchi (2025). “StitchFlow: Enabling In-Situ Creative Explorations of Crochet Patterns With Stitch Tracking and Process Sharing”. In: *Proceedings of the 38th Annual ACM Symposium on User Interface Software and Technology*. **UIST '25**. DOI: [10.1145/3746059.3747715](https://doi.org/10.1145/3746059.3747715).
- Marciniak, Zofia**, Kongpyung Moon, and Andrea Bianchi (2024). “Decoupling Geometry from Surface Finish by Parameterizing Texture Directly in G-code for Fused Deposition Modeling (FDM) Printing”. In: *Archives of Design Research* vol. 37. DOI: [10.15187/adr.2024.05.37.2.7](https://doi.org/10.15187/adr.2024.05.37.2.7).
- Moon, Kongpyung (Justin), **Zofia Marciniak**, Ryo Suzuki, and Andrea Bianchi (2024). “3D Printing Locally Activated Visual-Displays Embedded in 3D Objects via Electrically Conductive and Thermochromic Materials”. In: *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems*. **CHI '24**. Honolulu, HI, USA: Association for Computing Machinery. DOI: [10.1145/3613904.3642537](https://doi.org/10.1145/3613904.3642537).
- Marciniak, Zofia**, Kongpyung Moon, and Andrea Bianchi (2023). “3D Printing Slicer for Computational Texture Generation”. In: *Korea HCI '23*.
- Marciniak, Zofia**, Seo Young Oh, and Sang Ho Yoon (2022). “Guide Ring: Bidirectional Finger-worn Haptic Actuator for Rich Haptic Feedback”. In: *Proceedings of the 28th ACM Symposium on Virtual Reality Software and Technology*. **VRST '22**. Tsukuba, Japan. DOI: [10.1145/3562939.3565626](https://doi.org/10.1145/3562939.3565626).
- Marciniak, Zofia**, Sumin Han, and Dongman Lee (2021). “Understanding the impact of the Weather on Human Mobility via LTE Access Traces in Seoul Districts”. In: *Korea HCI '21*.

SELECTED RESEARCH GRANTS

- **SpaceTop: Spatial Computing HCI Technology for XR Productivity**
Sub-Project Manager: Presented quarterly updates, SIGCHI Korea posters, annual reports, and documentation.
- **3D-Printed Thermoplastic Volumetric Displays**
Project Manager: Authored final reports, managed documentation, created supporting media.
- **Electromagnetic Haptic Interface for Metaverse Interaction**
Lead Researcher: Proposed novel interaction system; presented at KAIST URP workshop; delivered final reports.
- **NH Investment & Securities–KAIST UX Research Center**
Project Participant: Conducted thematic analysis on investment behavior in metaverse environments.

SERVICE

- External Reviewer, ACM Designing Interactive Systems (DIS), 2026
- External Reviewer, ACM Symposium on User Interface Software and Technology (UIST), 2026

TALKS

- Guest Lecturer, Software Prototyping (ID311) (Spring 2025)
- Invited Speaker, KAIST's Wednesday Leadership Lecture (Fall 2026)

AWARDS & RECOGNITION

- iF Design Award — Evergrow: Investment Education Service, 2024
- Excellence in Leadership Award, KAIST, 2023